

METROPOLITAN TEACHING HOSPITAL & HEALTH SYSTEM

Diagnostic Microbiology & Inpatient Urology Division

NABH & NABL ACCREDITED

URINE CULTURE, AST & MEDICATION ORDER RECONCILIATION

REF: UC-2026-64112

Patient ID / Alias: PT-1039 (Ward 3 Bed 14)

Age / Gender: 54 Y / Female

Location: Ward 3B (General Inpatient)

Referring Dept: Urology & General MedicineCollection Date: 10-Sep-2026 11:15 AM

Allergy Status: Penicillin (Unspecified reaction)Reporting Date: 10-Sep-2026 09:30 AM

Current Antibiotic: Piperacillin-Tazobactam 4.5g IV TDS (Day 6 Timeout)

SPECIMEN & MICROBIOLOGICAL IDENTIFICATION

Specimen Type: Mid-Stream Clean Catch Urine (Cultured on Patient)Klebsiella pneumoniae (>10⁵ CFU/mL)

Direct Smear: Moderate pus cells (25-30/HPF), Gram-negative Colony Count: Significant bacteriuria (>100,000 CFU/mL)

ANTIMICROBIAL SUSCEPTIBILITY TESTING (CLSI M100 / EUCAST CRITERIA)

ANTIMICROBIAL AGENT	WHO AWARE	MIC (ug/mL)	INTERPRETATION
Nitrofurantoin	Access Tier (Narrow PO)	<=16	SUSCEPTIBLE
Ertapenem	Watch Tier	<=0.5	SUSCEPTIBLE
Amikacin	Access Tier	<=4.0	SUSCEPTIBLE
Ampicillin	Access Tier	>=32	RESISTANT
Trimethoprim-Sulfamethoxazole	Access Tier	>=4/76	RESISTANT
Ceftriaxone	Watch Tier	>=16	RESISTANT

BIOCHEMICAL & ORGAN FUNCTION BIOMARKERS

Serum Creatinine 1.1 mg/dL (Normal)	Blood Urea Nitrogen 16 mg/dL	Urine Pus Cells 25-30 / HPF	Oral Intake Tolerating liquids/solids
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ANTIMICROBIAL STEWARDSHIP CLINICAL SUMMARY & ACTION PLAN

TARGETED RECOMMENDATION: Step down to Nitrofurantoin 100mg PO QID for 5 to 7 days.

Clinical Rationale: Patient is clinically stable, afebrile, and tolerating oral nutrition. AST confirms K. pneumoniae susceptibility to narrow-spectrum agents.

Safety & Monitoring: Ensure patient renal clearance remains >30 mL/min (creatinine 1.1 mg/dL is safe for Nitrofurantoin concentration in low renal function).

Stewardship & Economic Impact: Enables early IV-to-PO switch, WHO Access tier compliance, removes central line risk, and reduces antibiotic costs.